ICR 26

Mini-project

06.05.2026

- AIs are **forbidden** for the **design phase**. You can use them to help you code and to help you improve the report.
- Submit **your code** and **your report** on Moodle.
- Describe the **key sizes, parameters, IVs, ...** in your report.
- Fix a **security level** and stick to it (e.g. 256-bit security).
- The programming language is free (preferably among Rust, C/C++, Java, Python/Sage). If you would like to use another language, please ask first.
- Do not hesitate to ask questions.
- The project has to be submitted **latest** on the **11.06** at 23h59.

1 A post-quantum ransomware

The goal of this project is to design a ransomware that is resistant to quantum computer.

- Victims should be able to unlock the ransomware from a simple password of 8-15 characters taken from a dictionary (you can hardcode the dictionary for simplicity).
- If someone reverse-engineers the malware, he should not be able to decrypt.
- Once a user pays, he receives the password to decrypt everything.
- It should be possible to ask for the decryption of a specific file (and to pay a lesser fee). In this case, he doesn't receive a password but a symmetric key.
- The ransomware should be post-quantum.
- You should maximize the success rate of your ransomware. In particular, you should make it useable offline (as much as possible). For instance, your ransomware could be executed on a laptop while in the plane and obtain network connection just after the flight. We can also assume that the network is cut after a suspicious connection.
- To simplify, you can assume that the ransomware encrypts only the files inside a folder (without any depth).
- Once paid, the ransomware should be sure that the received password is the legitimate one and was not modified on the way by a third party. Something similar should happen when unlocking a single file.
- You can implement the following testing interface:
  1. encrypt (encrypts all the files)
  2. Pay (recovers the global password)
  3. Unlock one file (lists files and ask which one to unlock for a smaller fee)

2 Deliverable

You have to deliver the following:

- A report describing your cryptographic architecture and explaining your choices. In particular, provide a figure describing how the keys are managed. Explain also why it is resistant to quantum attacks. Justify why you have the same security level throughout your application.
- Your code. Note that we do **not** ask you to implement any networking. If you want, you can simulate everything locally. Only the cryptographic part will be evaluated.
- Bonus points will be given for any cool (mostly crypto-related) additional functionality. **Describe them in your report so that I know what you did!**.
- Explain clearly how you used AI, if used.